



VMWARE VSPHERE 7.0: INSTALL, CONFIGURE, MANAGE



Lab 04: Configuring and Deploying a Virtual Machine

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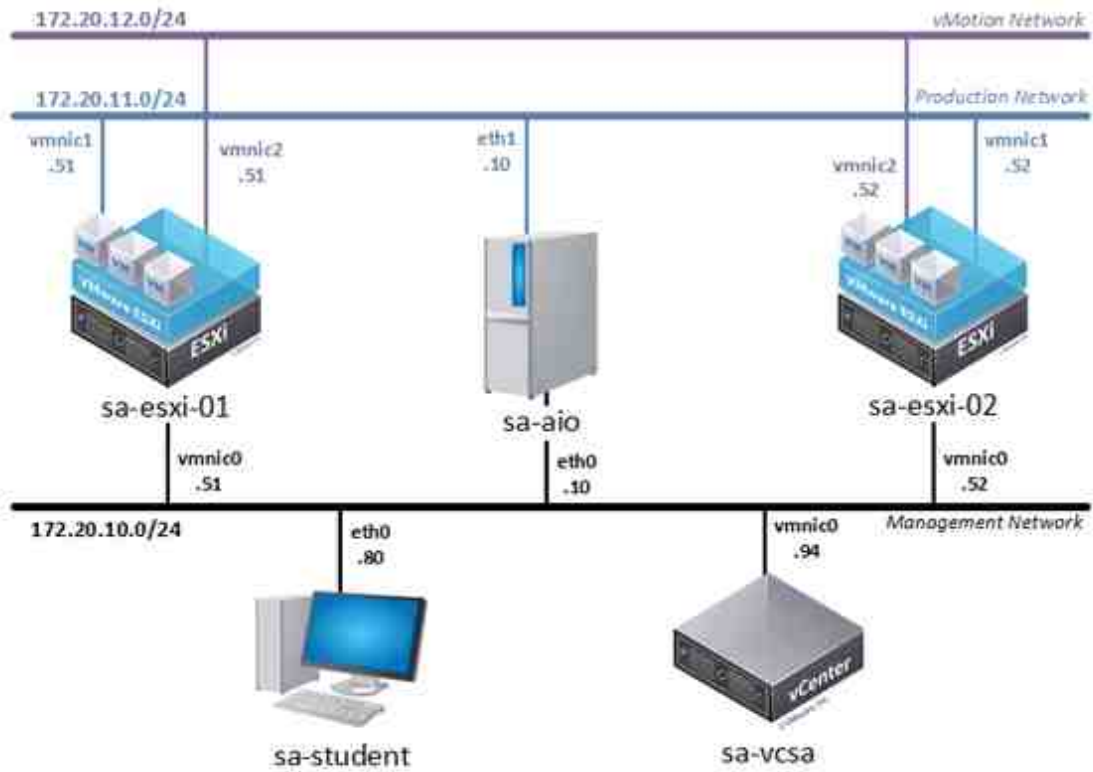
Introduction

In this lab, you will create and prepare a virtual machine for use.

Objectives

-) Create a virtual machine
-) Install a guest operating system
-) Install VMware Tools
-) Uninstall VMware Tools

Lab Topology



Lab Settings

The information in the table below will be needed in order to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account	Password
sa-esxi-01	eth0: 172.20.10.51	root	Train1ng\$
sa-esxi-02	eth0: 172.20.10.52	root	Train1ng\$
sa-aio	eth0: 172.20.10.10	sysadmin	vmware123
sa-student	eth0: 172.20.10.80	sysadmin	vmware123
sa-vcasa	eth0: 172.20.10.94	sysadmin@vclass.local	vmware123

1 Create a Virtual Machine

In this task, you will create a virtual machine based on specific requirements, such as a particular operating system or hardware configuration.

1. Launch the sa-student virtual machine to access the graphical login screen.

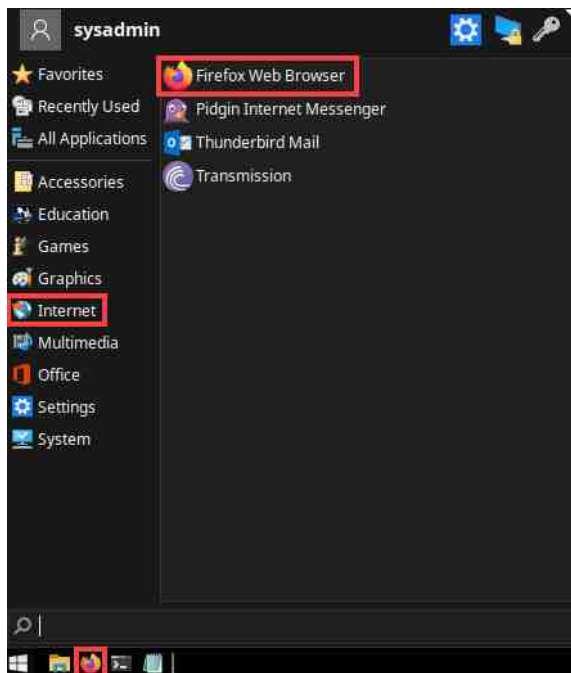


To launch the console window for a virtual machine, you may access by either clicking on the machine's graphic image from the topology page or by clicking on the machine's respective tab from the navigation bar.

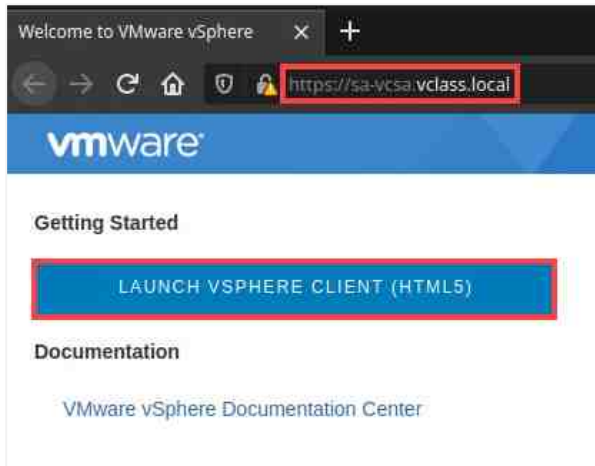
2. Log in as `sysadmin` using the password `vmware123`.



3. Launch the Mozilla Firefox web browser by either clicking on the icon shortcut found on the bottom toolbar or by navigating to Start Menu > Internet > Firefox Web Browser.

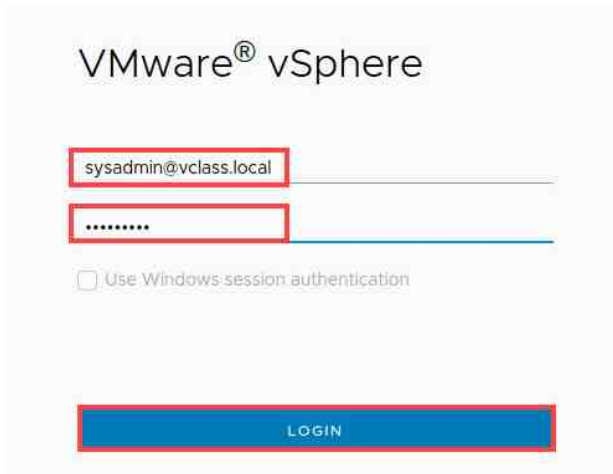


4. Notice the homepage is automatically configured to load the URL address of the VMware vCenter Server Appliance (sa-vcasa virtual machine). Click on the LAUNCH VSPHERE CLIENT (HTML5) button to launch the vSphere Client.



If you cannot successfully load the VMware Getting Started webpage, then please wait an additional 3-5 minutes and refresh the page to continue. During this time, the vCenter Server Appliance is still booting up and requires extra time to initialize.

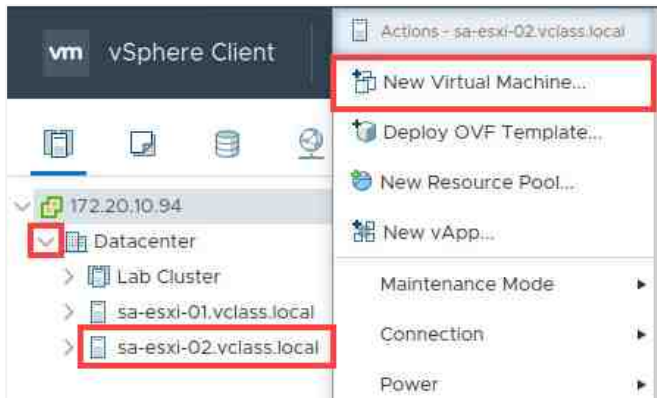
5. In the Username text field, type `sysadmin@vclass.local` and in the Password text field, type `vmware123`. Click on LOGIN.



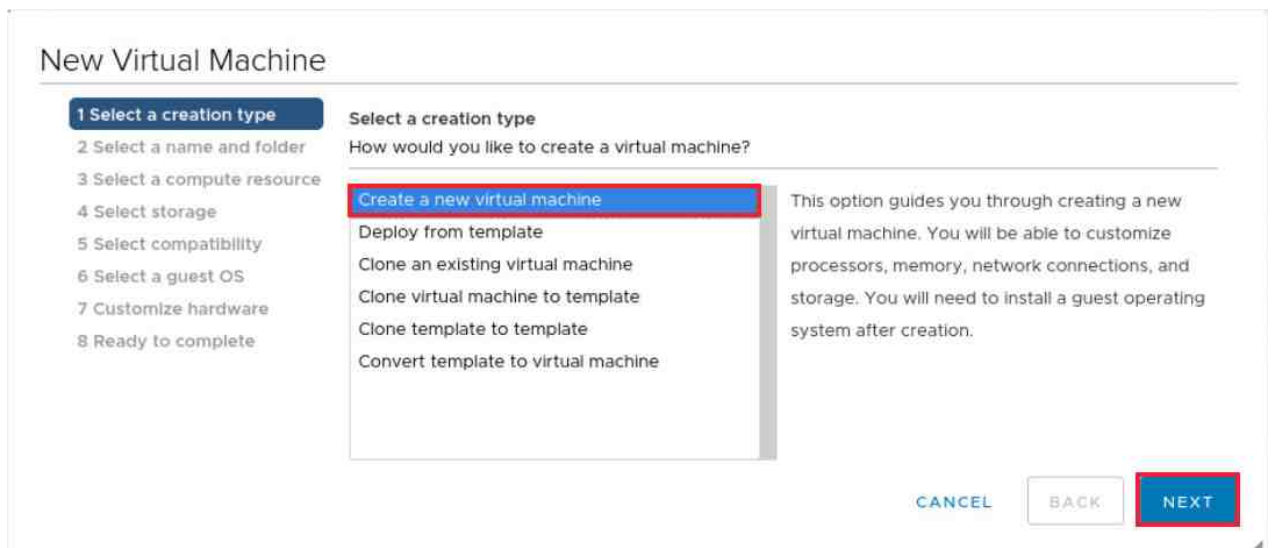
**Please
Note**

You may ignore the “browser-OS combination” warning message presented on the VMware vCenter Single Sign-On page and continue moving forward with the lab.

- Expand the Datacenter object in the navigator pane. Right-click on sa-esxi-02.vclass.local and select New Virtual Machine.



- In the New Virtual Machine window, on the Select a creation type step, select Create a new virtual machine and click Next.



- On the Select a name and folder step, enter VM2-2 as the name and ensure that Datacenter is selected for the location. Click Next.

New Virtual Machine

✓ 1 Select a creation type

2 Select a name and folder

3 Select a compute resource

4 Select storage

5 Select compatibility

6 Select a guest OS

7 Customize hardware

8 Ready to complete

Select a name and folder

Specify a unique name and target location

Virtual machine name:

Select a location for the virtual machine.

✓ 172.20.10.94

> **Datacenter**

CANCEL

BACK

NEXT

- On the Select a compute resource step, ensure that sa-esxi-02.vclass.local is selected and click Next.

New Virtual Machine

✓ 1 Select a creation type

✓ 2 Select a name and folder

3 Select a compute resource

4 Select storage

5 Select compatibility

6 Select a guest OS

7 Customize hardware

8 Ready to complete

Select a compute resource

Select the destination compute resource for this operation

✓ Datacenter

> Lab Cluster

> sa-esxi-01.vclass.local

> **sa-esxi-02.vclass.local**

Compatibility

✓ Compatibility checks succeeded.

CANCEL

BACK

NEXT

10. On the Select storage step, select Local02-2 to be the local datastore and click Next.

New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- 4 Select storage**
- 5 Select compatibility
- 6 Select a guest OS
- 7 Customize hardware
- 8 Ready to complete

Select storage
Select the storage for the configuration and disk files

☐ Encrypt this virtual machine (Requires Key Management Server)

VM Storage Policy: Datastore Default

Name	Capacity	Provisioned	Free	Type	Cluster
datastore 2	14.75 GB	24.03 GB	9.79 GB	VMFS 6	
ISCSH-LUN0	35 GB	12.29 GB	22.71 GB	VMFS 5	
Local02-2	14.75 GB	1.41 GB	13.34 GB	VMFS 6	
NFS-1	9.62 GB	0 B	9.62 GB	NFS v3	

Compatibility
✓ Compatibility checks succeeded.

CANCEL BACK NEXT

11. On the Select compatibility step, select ESXi 7.0 and later. Click Next.

New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Select storage
- 5 Select compatibility**
- 6 Select a guest OS
- 7 Customize hardware
- 8 Ready to complete

Select compatibility
Select compatibility for this virtual machine depending on the hosts in your environment

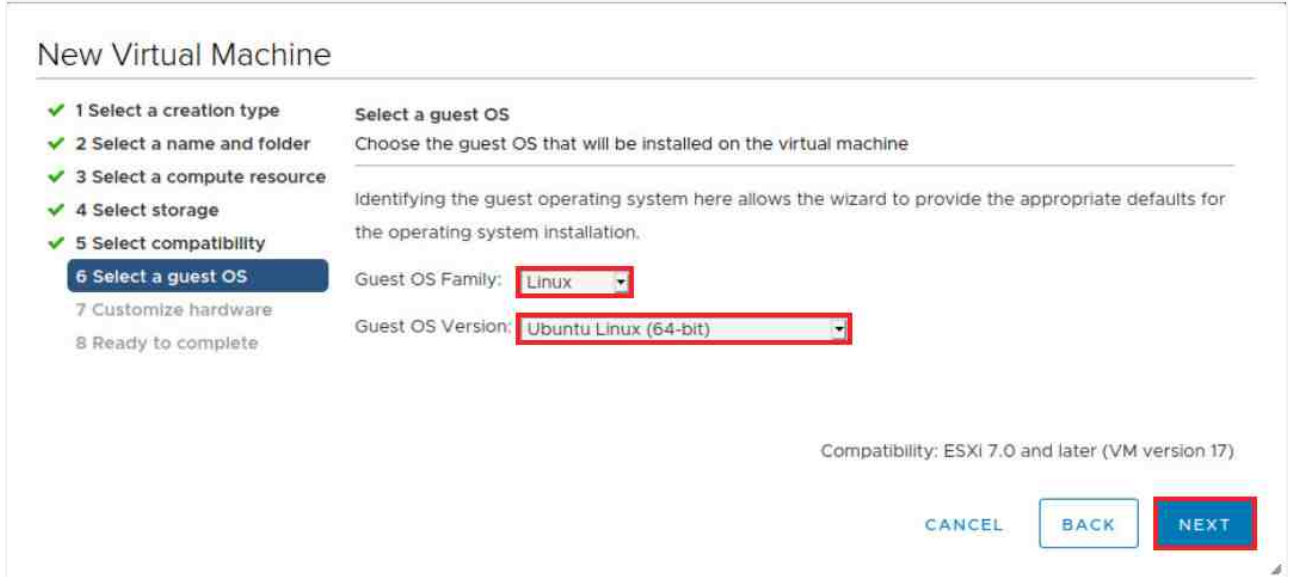
The host or cluster supports more than one VMware virtual machine version. Select a compatibility for the virtual machine.

Compatible with: **ESXi 7.0 and later** ⓘ

This virtual machine uses hardware version 17, which provides the best performance and latest features available in ESXi 7.0

CANCEL BACK NEXT

12. On the Select a guest OS step, select Linux as the Guest OS Family and Ubuntu Linux (64-bit) as the Guest OS Version. Click Next.



New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Select storage
- ✓ 5 Select compatibility
- 6 Select a guest OS**
- 7 Customize hardware
- 8 Ready to complete

Select a guest OS
Choose the guest OS that will be installed on the virtual machine

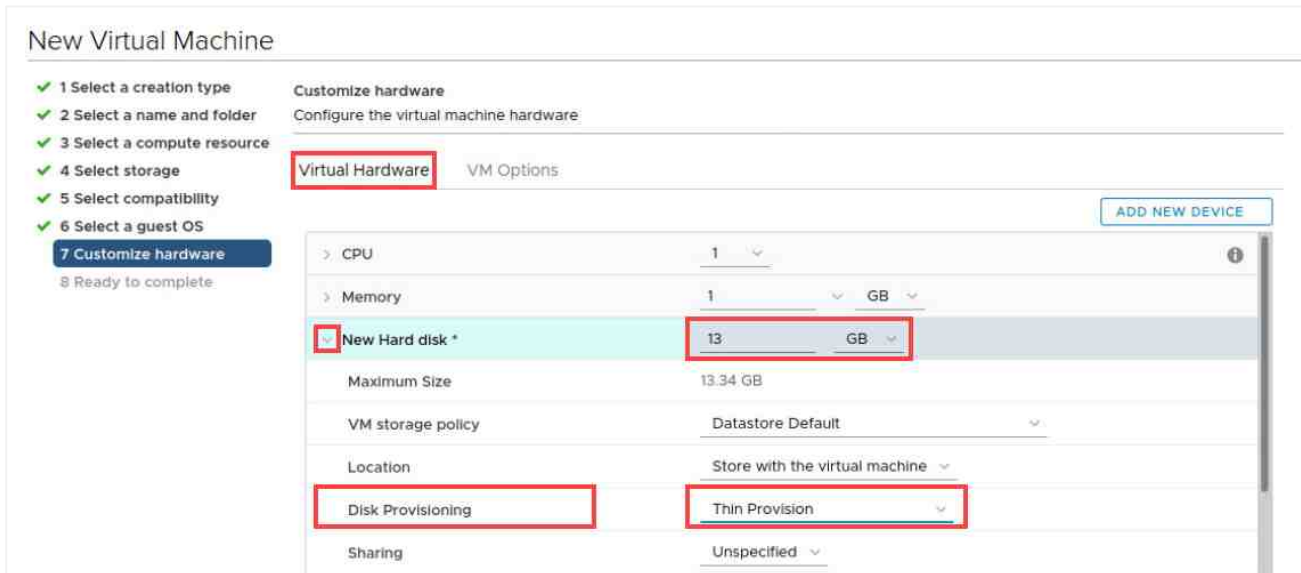
Identifying the guest operating system here allows the wizard to provide the appropriate defaults for the operating system installation.

Guest OS Family: Linux
Guest OS Version: Ubuntu Linux (64-bit)

Compatibility: ESXi 7.0 and later (VM version 17)

CANCEL BACK NEXT

13. On the Customize hardware, Virtual Hardware tab, expand New Hard disk and enter 13 GB for the hard disk size and select Thin Provision for the Disk Provisioning.



New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Select storage
- ✓ 5 Select compatibility
- ✓ 6 Select a guest OS
- 7 Customize hardware**
- 8 Ready to complete

Customize hardware
Configure the virtual machine hardware

Virtual Hardware VM Options

ADD NEW DEVICE

CPU 1

Memory 1 GB

New Hard disk * 13 GB

Maximum Size 13.34 GB

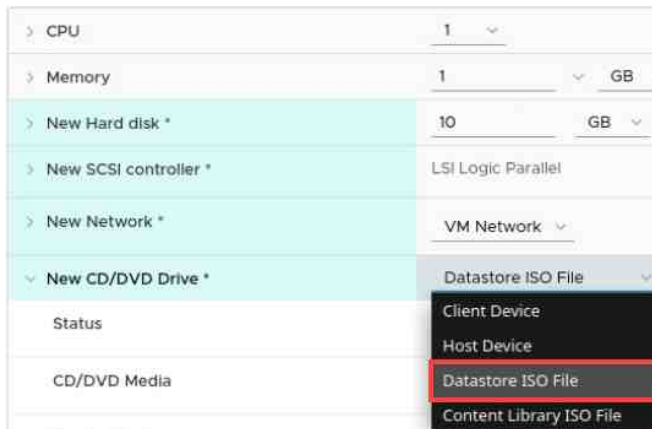
VM storage policy Datastore Default

Location Store with the virtual machine

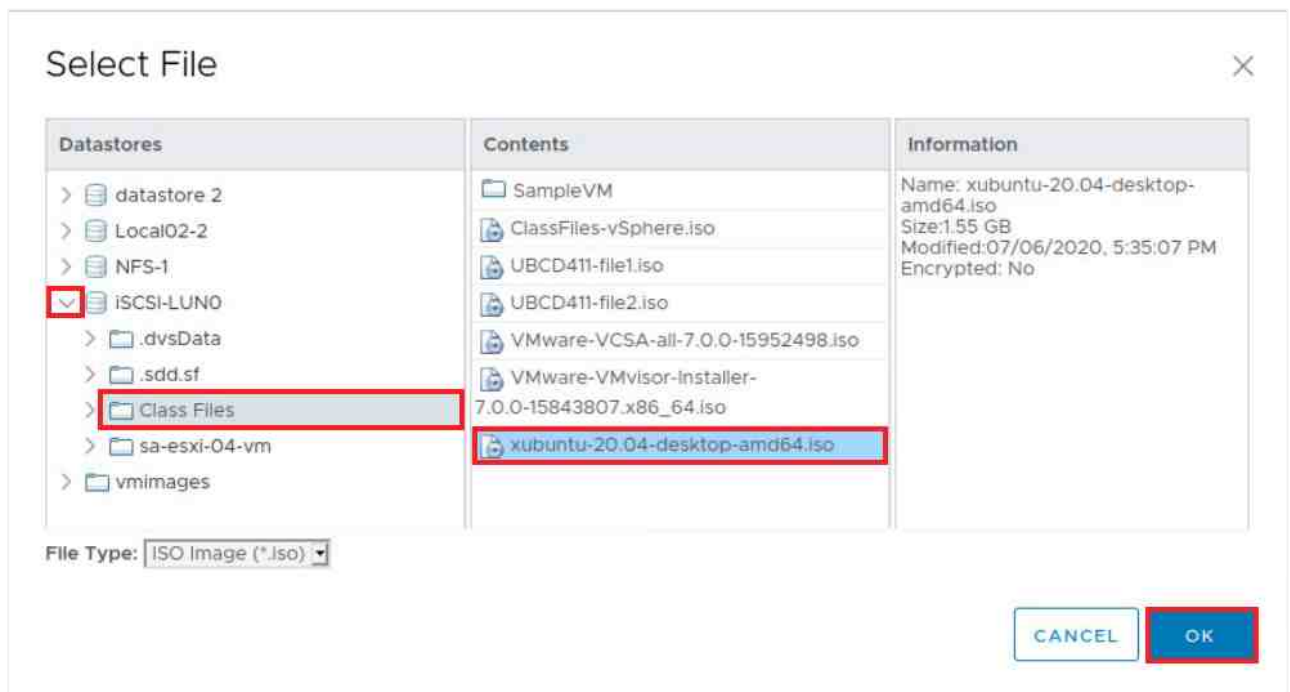
Disk Provisioning Thin Provision

Sharing Unspecified

14. Locate the New CD/DVD Drive. Select Datastore ISO File from the dropdown menu.



15. In the Select File dialog box, navigate to iSCSI-LUN0 > Class Files underneath the Datastores column and then proceed by selecting xubuntu-20.04-desktop-amd64.iso underneath the Contents column. Click OK.



16. Back on the Customize hardware step, expand the New CD/DVD Drive. Check the checkbox for Connect At Power On. Click Next.

New Virtual Machine

1 Select a creation type
2 Select a name and folder
3 Select a compute resource
4 Select storage
5 Select compatibility
6 Select a guest OS
7 **Customize hardware**
8 Ready to complete

Customize hardware
Configure the virtual machine hardware

Virtual Hardware VM Options

ADD NEW DEVICE

> CPU	1	
> Memory	1	GB
> New Hard disk *	13	GB
> New SCSI controller *	LSI Logic Parallel	
> New Network *	VM Network	☒ Connect...
☒ New CD/DVD Drive *	Datastore ISO File	
Status	☒ Connect At Power On	
CD/DVD Media	[iSCSI-LUN0] Class Files/xubur BROWSE...	
Device Mode	Passthrough CD-ROM	
Virtual Device Node	New SATA Controller	SATA(0:0) New CD/DVD Drive
> Video card *	Specify custom settings	

Compatibility: ESXi 7.0 and later (VM version 17)

CANCEL BACK **NEXT**

17. On the Ready to complete step, review the information and click Finish.

New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Select storage
- ✓ 5 Select compatibility
- ✓ 6 Select a guest OS
- ✓ 7 Customize hardware
- 8 Ready to complete**

Ready to complete
Click Finish to start creation.

Virtual machine name	VM2-2
Folder	Datacenter
Host	sa-esxi-02.vclass.local
Datastore	Local02-2
Guest OS name	Ubuntu Linux (64-bit)
Virtualization Based Security	Disabled
CPUs	1
Memory	1 GB
NICs	1
NIC 1 network	VM Network
NIC 1 type	VMXNET 3
SCSI controller 1	LSI Logic Parallel
Create hard disk 1	New virtual disk
Capacity	13 GB
Datastore	Local02-2

CANCEL BACK **FINISH**

18. Notice the action event in the Recent Tasks pane. Also, take notice that the VM2-2 now appears underneath sa-esxi-02 in the navigator pane. Select VM2-2 to view more information.

vm vSphere Client

Menu Search in all environments

172.20.10.94 Datacenter Lab Cluster sa-esxi-01.vclass.local **sa-esxi-02.vclass.local** Linux-02 Photon-02 sa-esxi-04-vm **VM2-2**

VM2-2

Summary Monitor Configure Permissions Datastores Networks Snapshots Updates

CUSTOMIZE VIEW

Guest OS

Power Status: Powered Off

Guest OS: Ubuntu Linux (64-bit)

Encryption: Not encrypted

VMware Tools: Not running, not installed ⓘ

DNS Name: --

IP Addresses: --

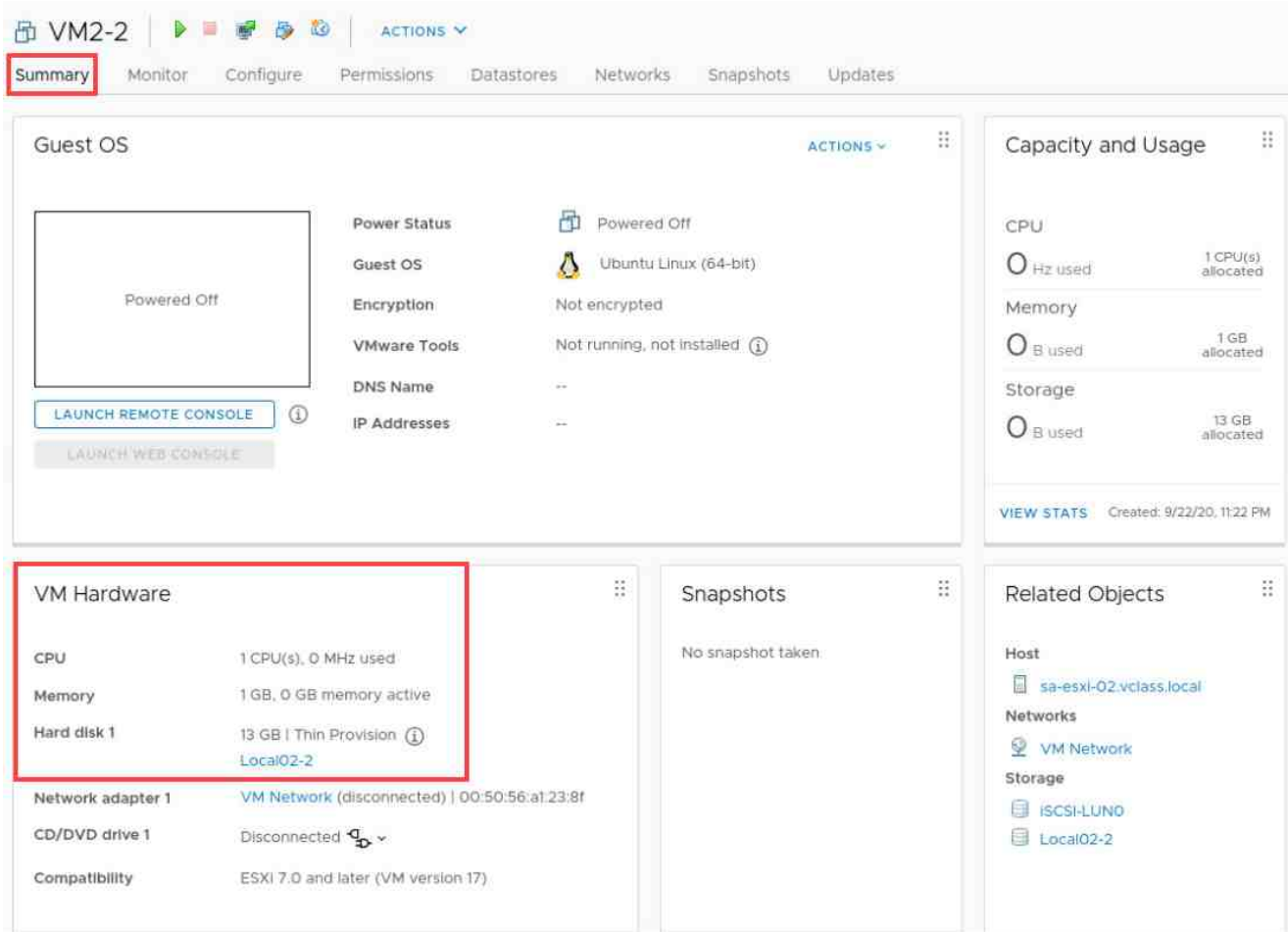
LAUNCH REMOTE CONSOLE ⓘ

LAUNCH WEB CONSOLE

Recent Tasks

Task Name	Target	Status	Details	Initiator	Queued For	Start Time	Completion Time	Server
Create virtual machine	Datacenter	✓ Completed	Copying Virtual Machine configuration	VCLASS.LOCAL\sys...	10 ms	07/21/2020, 4:57:26 PM	07/21/2020, 4:57:28 PM	172.20.10.94

19. In the main workspace pane, ensure that you are viewing the Summary tab of VM2-2. On the VM Hardware panel, verify the CPU, Memory, and Hard Disk1 size is correct and set to Thin Provision. Lastly, verify that VM2-2 has been installed on Local02-2.



The screenshot displays the vSphere Client interface for VM2-2. The 'Summary' tab is selected and highlighted with a red box. The VM is in a 'Powered Off' state. The 'VM Hardware' section, also highlighted with a red box, lists the following specifications:

- CPU:** 1 CPU(s), 0 MHz used
- Memory:** 1 GB, 0 GB memory active
- Hard disk 1:** 13 GB | Thin Provision (Local02-2)
- Network adapter 1:** VM Network (disconnected) | 00:50:56:a1:23:8f
- CD/DVD drive 1:** Disconnected
- Compatibility:** ESXi 7.0 and later (VM version 17)

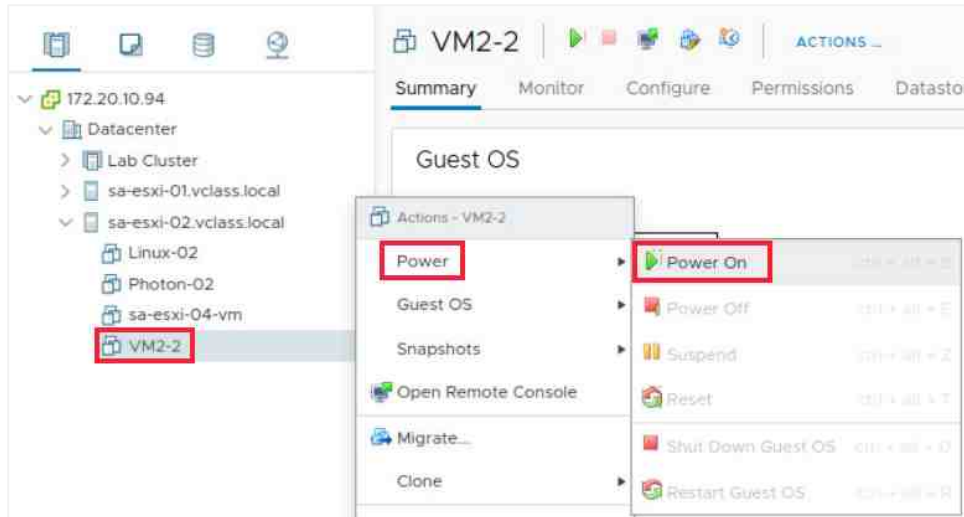
Other visible sections include 'Guest OS' (Ubuntu Linux (64-bit)), 'Capacity and Usage' (1 CPU(s) allocated, 1 GB allocated, 13 GB allocated), 'Snapshots' (No snapshot taken), and 'Related Objects' (Host: sa-esxi-02.vclass.local, Networks: VM Network, Storage: iSCSI-LUN0, Local02-2).

20. Leave the vSphere Client opened to continue with the next task.

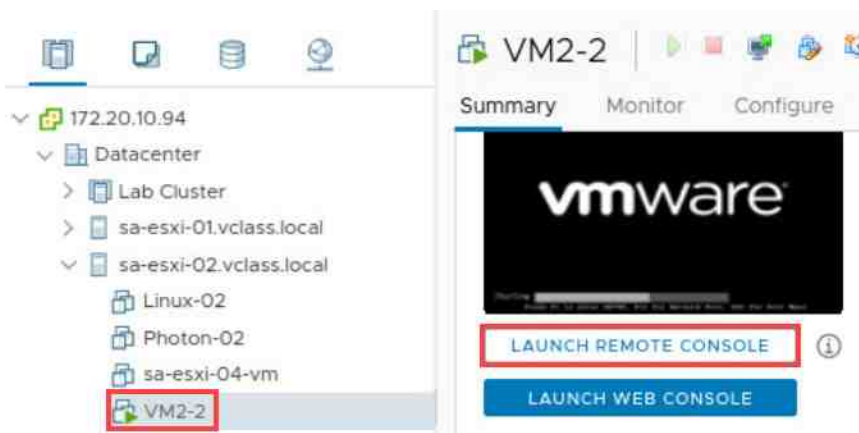
2 Install a Guest Operating System

In this task, you will install an operating system.

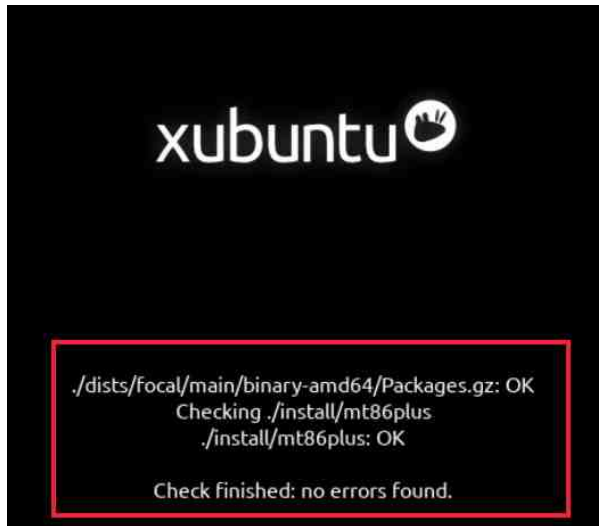
1. While in the vSphere Web Client, right-click on VM2-2 in the navigator pane and select Power > Power On.



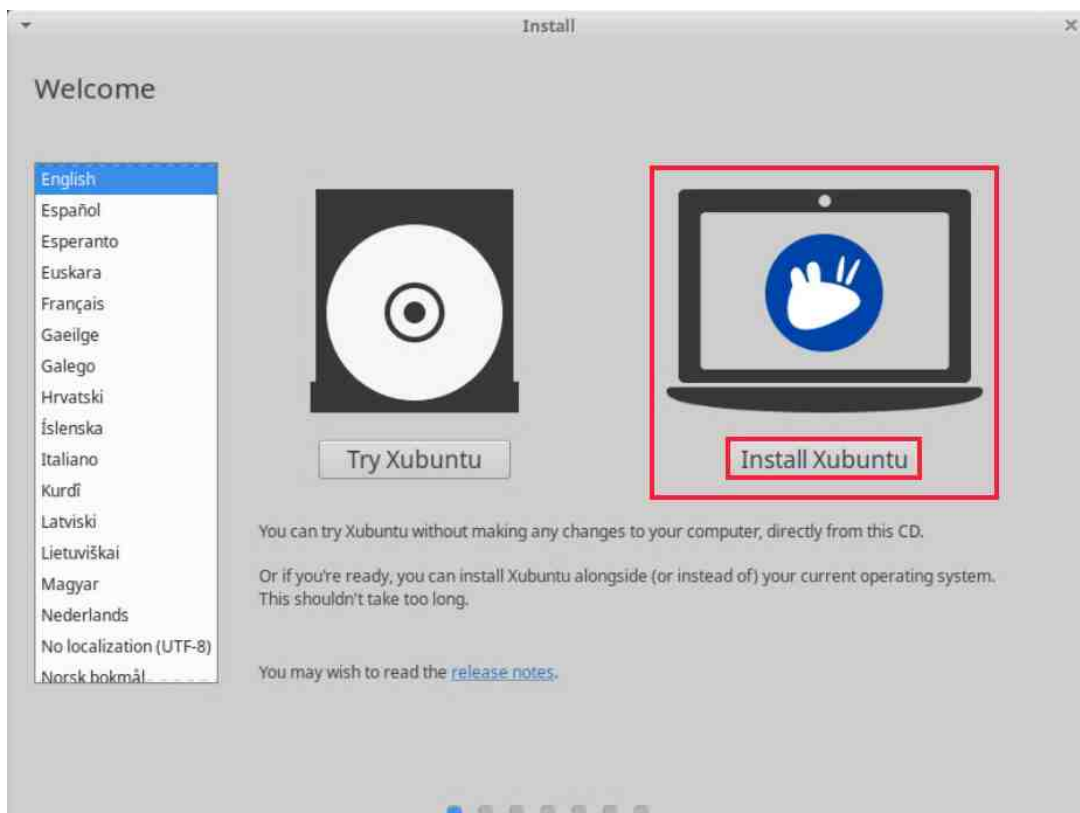
2. Verify that VM2-2 is selected and click LAUNCH REMOTE CONSOLE.



3. In the console, if a fragment check is in process, you may cancel the request by pressing the Esc key. You may also let this process complete and proceed to the next step.



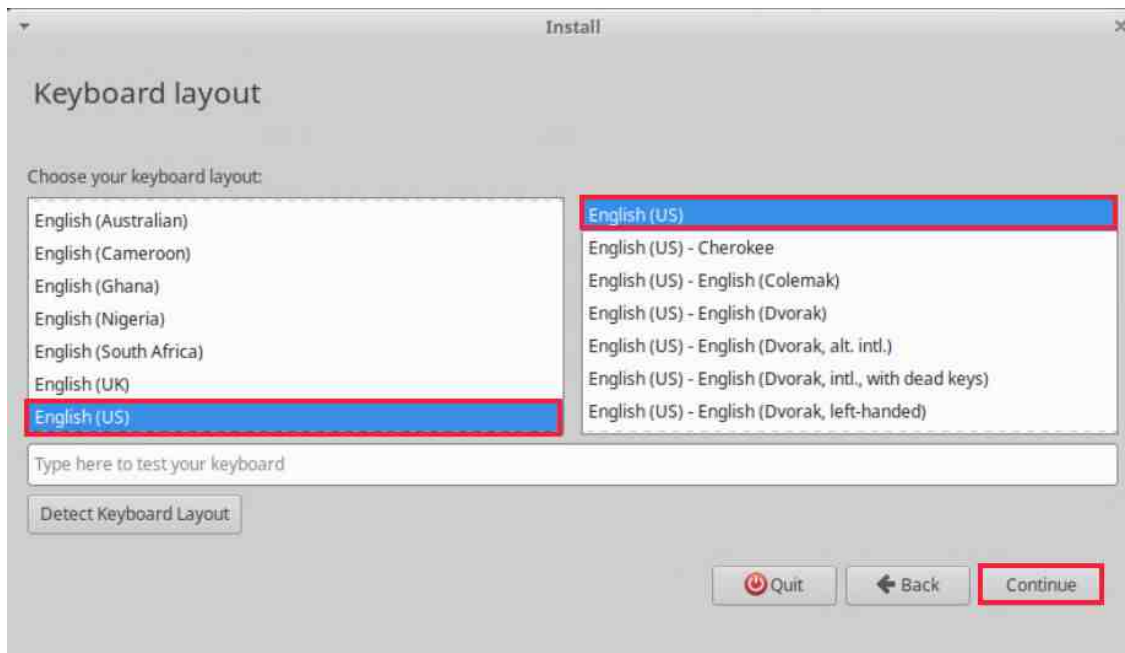
4. Once the Xubuntu Install dialog box appears, click on the Install Xubuntu graphic.



**Please
Note**

If the Xubuntu installation wizard does not appear immediately, wait 1-2 minutes for the wizard to load.

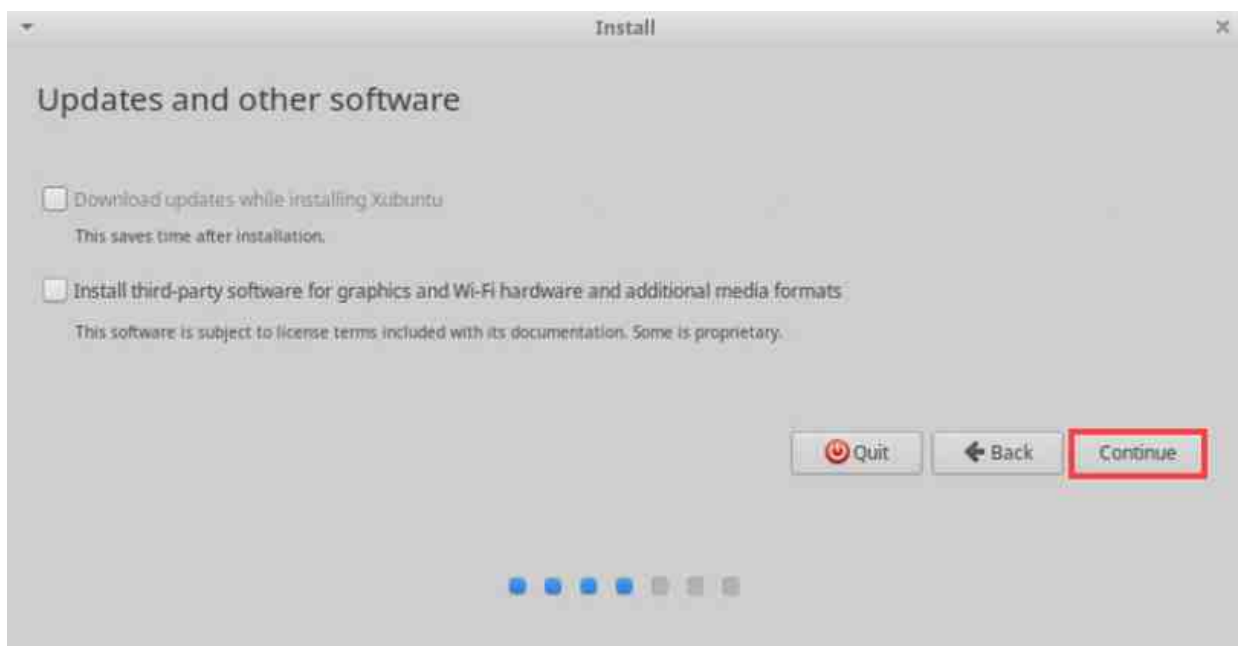
- Once the Keyboard layout window appears, select English (US) > English (US). Click Continue.



**Please
Note**

For this lab example, we chose to use English(US), but you may choose to use any keyboard type.

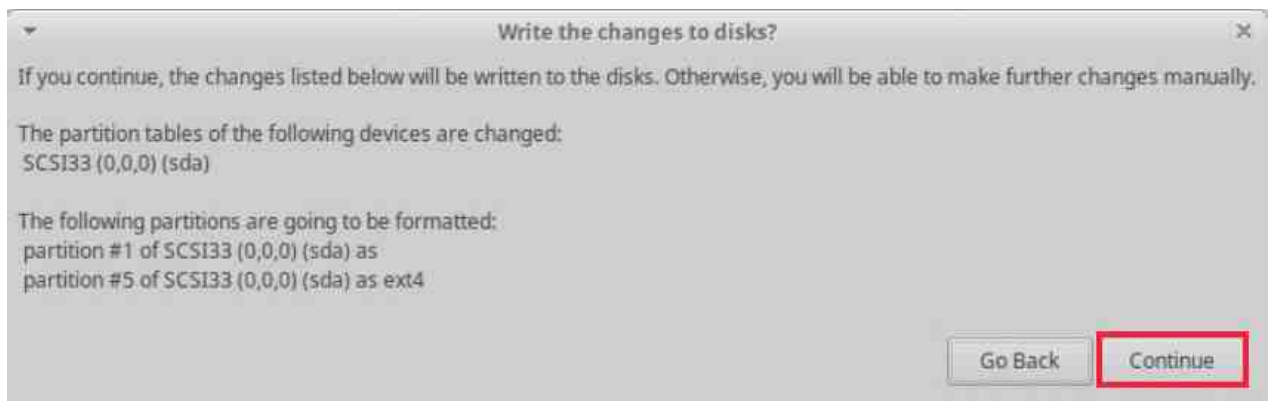
- On the Updates and other Software window, click Continue.



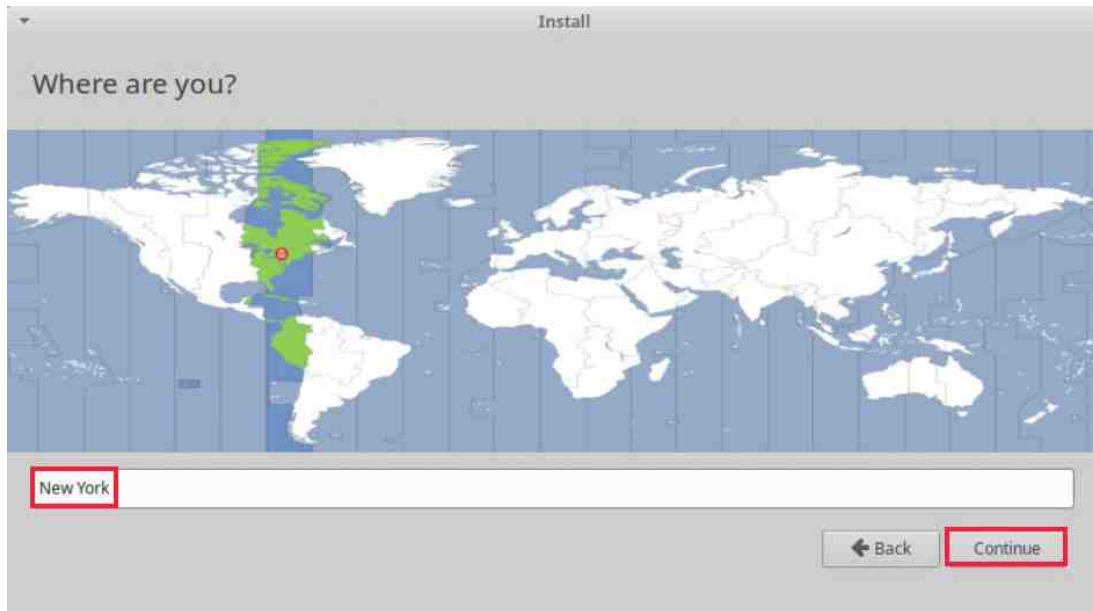
7. On the Installation type window, leave the defaults selected and click Install Now.



8. On the Write the changes to disks? window, click Continue.

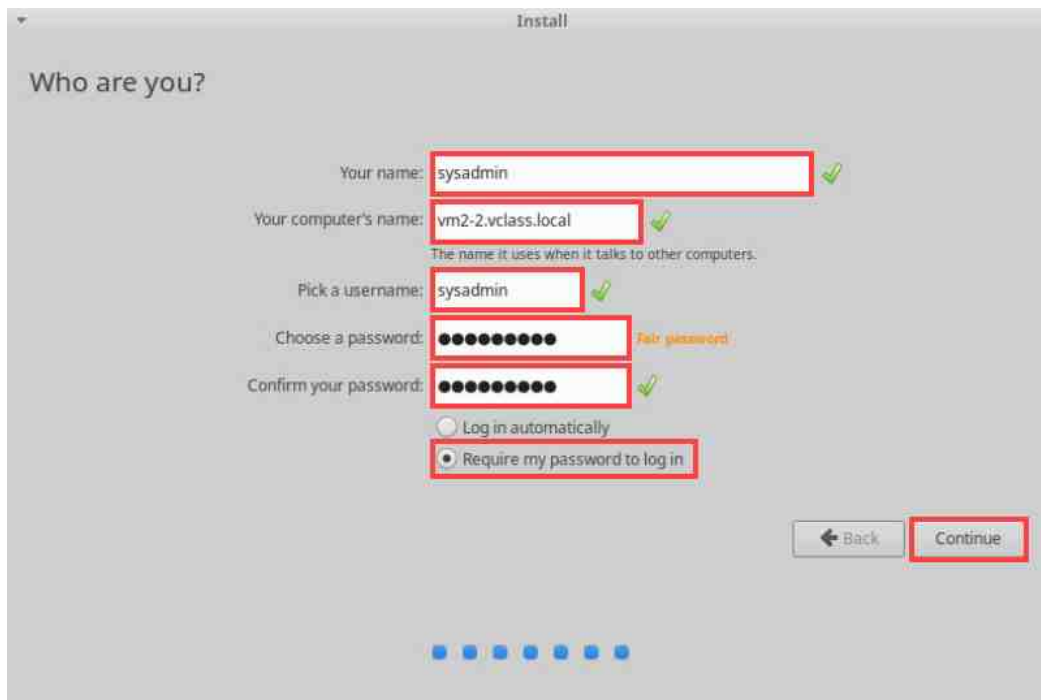


9. In the Where are you? window, verify New York is in the time zone field and click Continue.

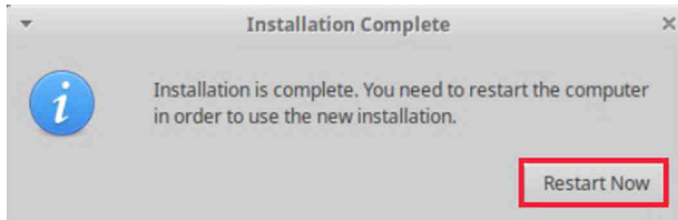
**Please Note**

For this lab example, we chose to use the New York time zone, but you may choose to use any time zone.

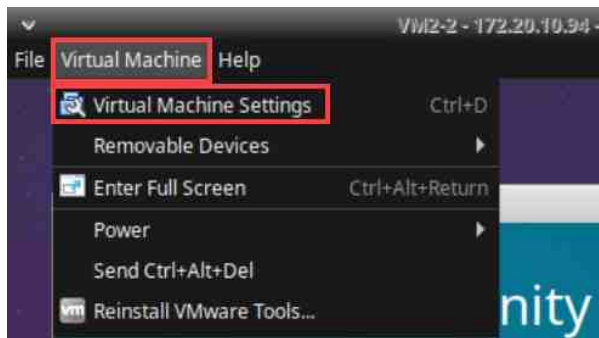
10. On the Who are you? window, enter `sysadmin` for Your Name, enter `vm2-2.vclass.local` for Your computer's name, enter `sysadmin` for Pick a username, enter `vmware123` in both the Choose a password and Confirm your password fields, verify Require my password to log in is selected, and lastly, click Continue.



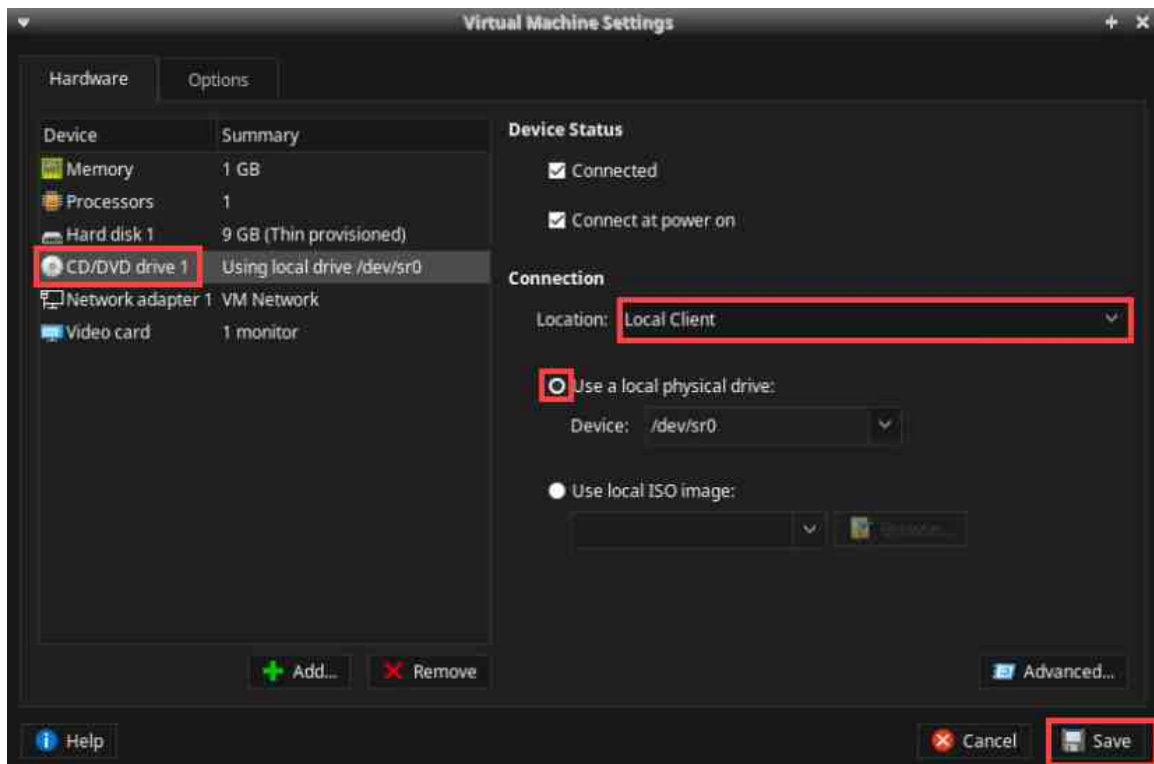
11. The installation process will take five to seven minutes to complete. Once the Installation Complete window appears, click Restart Now.



12. While rebooting, eject the installation CD from the virtual disk drive by clicking on the Virtual Machine menu option and selecting Virtual Machine Settings or by pressing CTRL+D as a shortcut.



13. In the Virtual Machine Settings window, select CD/DVD drive 1 from the Device pane and select Local Client from the Location dropdown menu, followed by selecting Use a local physical drive. Click Save.



14. Press Enter, verifying that the installation medium has been disconnected.

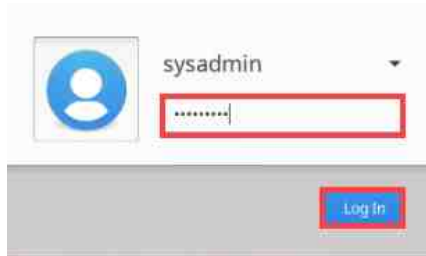


15. Leave the VM2-2 console open to continue with the next task.

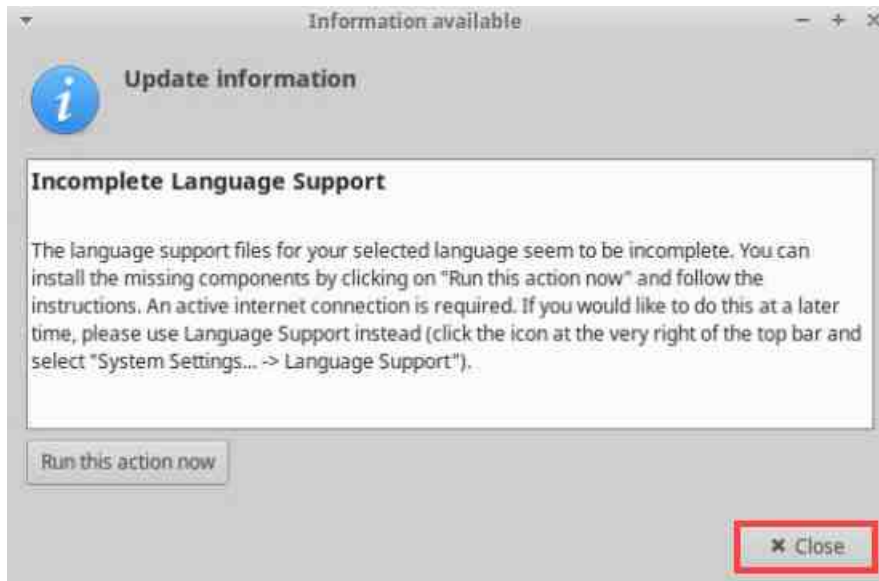
3 Install VMware Tools

In this task, you will install VMware tools via the command line.

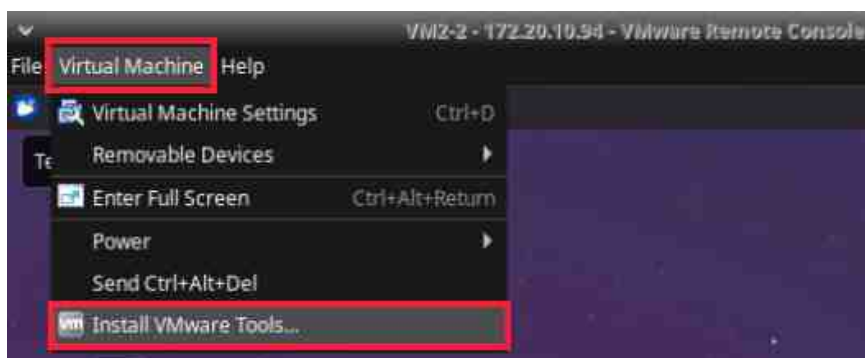
1. On VM2-2, enter `vmware123` in the password field and click Log In.



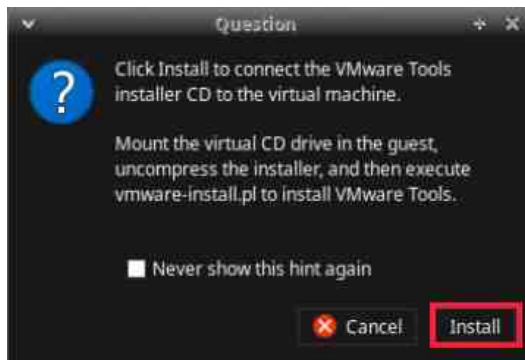
2. On the Information available window, click Close.



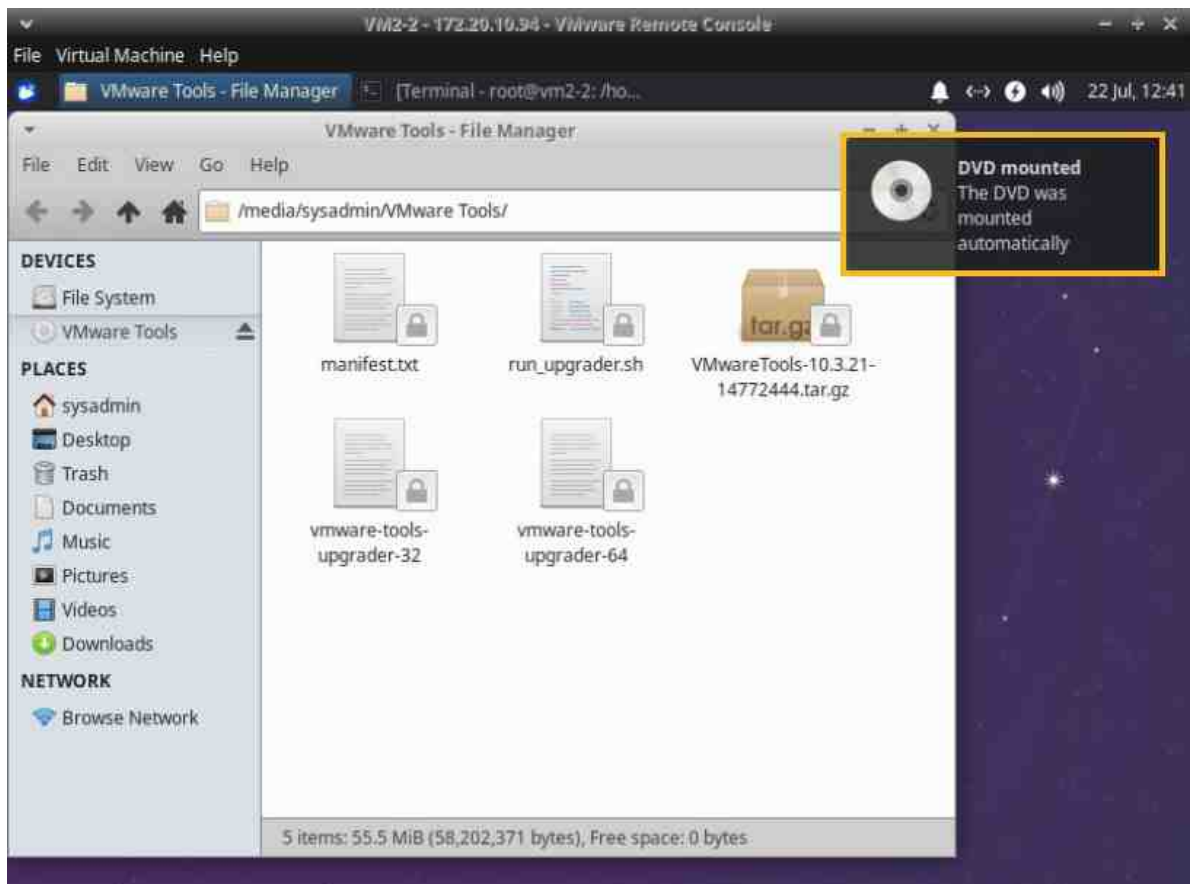
3. To begin the official VMware Tools install, click on the Virtual Machine menu option and click Install VMware Tools.



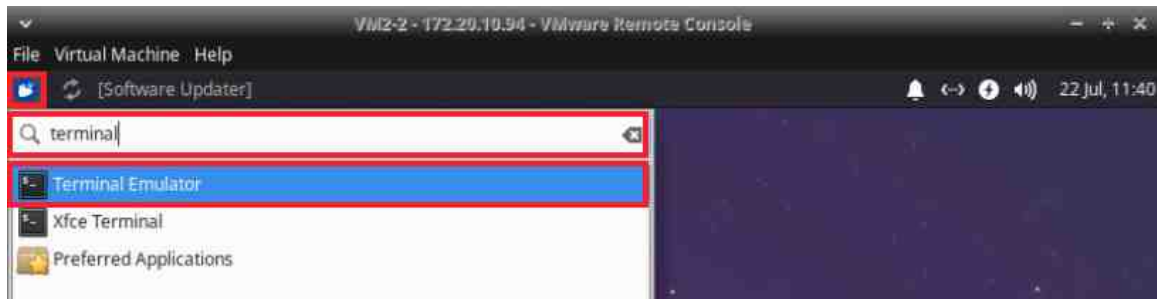
4. In the Question window, click Install to mount the virtual CD drive.



5. Notice VMware Tools was mounted and ready to install. Minimize the VMware Tools - File Manager window.



6. While consoled into VM2-2, click on the Whisker icon and type `terminal` in the search field. Click on Terminal Emulator to launch it.



7. Elevate your privileges by becoming the superuser of the system. When prompted for the root password, enter `vmware123` and press Enter.

```
sysadmin@vm2-2:~$ sudo su
```

```
sysadmin@vm2-2:~$ sudo su
[sudo] password for sysadmin:
root@vm2-2:/home/sysadmin#
```

8. In the terminal window, create a mounting directory.

```
root@vm2-2:/home/sysadmin# mkdir /mnt/cdrom
```

```
root@vm2-2:/home/sysadmin# mkdir /mnt/cdrom
root@vm2-2:/home/sysadmin#
```

9. Mount the CD to the directory.

```
root@vm2-2:/home/sysadmin# mount /dev/cdrom /mnt/cdrom
```

```
root@vm2-2:/home/sysadmin# mount /dev/cdrom /mnt/cdrom/
mount: /mnt/cdrom: WARNING: device write-protected, mounted read-only.
root@vm2-2:/home/sysadmin#
```

10. View the contents of the mounted CD ROM.

```
root@vm2-2:/home/sysadmin# ls /mnt/cdrom
```

```
root@vm2-2:/home/sysadmin# ls /mnt/cdrom/
manifest.txt  VMwareTools-10.3.21-14772444.tar.gz  vmware-tools-upgrader-64
run_upgrader.sh  vmware-tools-upgrader-32
root@vm2-2:/home/sysadmin#
```

11. Copy the tar.gz file to a temporary directory.

```
root@vm2-2:/home/sysadmin# cp /mnt/cdrom/*.gz /tmp
```

```
root@vm2-2:/home/sysadmin# cp /mnt/cdrom/*.gz /tmp
root@vm2-2:/home/sysadmin#
```

12. Change your current working directory to /tmp.

```
root@vm2-2:/home/sysadmin# cd /tmp
```

```
root@vm2-2:/home/sysadmin# cd /tmp/
root@vm2-2:/tmp#
```

13. View the contents of the /tmp directory and verify that the VMwareTools-10.3.21-14772444.tar.gz file is present.

```
root@vm2-2:/tmp# ls
```

```
root@vm2-2:/tmp# ls
config-err-MyZk5Q
ssh-071JWhP4iQP5
systemd-private-a3196863ac57449eb4795c39ffdf9ef8-colord.service-W47t0h
systemd-private-a3196863ac57449eb4795c39ffdf9ef8-ModemManager.service-PCDMDi
systemd-private-a3196863ac57449eb4795c39ffdf9ef8-systemd-logind.service-ZFxfFf
systemd-private-a3196863ac57449eb4795c39ffdf9ef8-systemd-resolved.service-qFCRKh
systemd-private-a3196863ac57449eb4795c39ffdf9ef8-systemd-timesyncd.service-SSw4eg
systemd-private-a3196863ac57449eb4795c39ffdf9ef8-upower.service-M6dvog
VMwareTools-10.3.21-14772444.tar.gz
```

14. Use TAR, the package manager, to unpack the VMwareTools-10.3.21-14772444.tar.gz file.

```
root@vm2-2:/tmp# tar -zxvf VMwareTools-10.3.21-14772444.tar.gz
```

```
root@vm2-2:/tmp# tar -zxvf VMwareTools-10.3.21-14772444.tar.gz
vmware-tools-distrib/lib/plugins64/vmusr/
vmware-tools-distrib/lib/plugins64/vmusr/libdndcp.so
vmware-tools-distrib/lib/plugins64/vmusr/libdesktopEvents.so
vmware-tools-distrib/lib/plugins64/vmusr/libresolutionSet.so
vmware-tools-distrib/lib/plugins64/common/
vmware-tools-distrib/lib/plugins64/common/libvix.so
vmware-tools-distrib/lib/plugins64/common/libhgfsServer.so
vmware-tools-distrib/doc/
vmware-tools-distrib/doc/INSTALL
vmware-tools-distrib/doc/open_source_licenses.txt
vmware-tools-distrib/doc/README
vmware-tools-distrib/vmware-install.pl
root@vm2-2:/tmp#
```

15. Navigate to the vmware-tools-distrib directory.

```
root@vm2-2:/tmp# cd vmware-tools-distrib
```

```
root@vm2-2:/tmp# cd vmware-tools-distrib
root@vm2-2:/tmp/vmware-tools-distrib#
```

16. View the contents of the vmware-tools-distrib and verify vmware-install.pl is listed.

```
root@vm2-2:/tmp/vmware-tools-distrib# ls
```

```
root@vm2-2:/tmp/vmware-tools-distrib# ls
bin  caf  doc  etc  FILES  INSTALL  installer  lib  vgauth  vmware-install.pl
root@vm2-2:/tmp/vmware-tools-distrib#
```

17. Execute the VMware Tools installation script.

```
root@vm2-2:/tmp/vmware-tools-distrib# ./vmware-install.pl
```

```
root@vm2-2:/tmp/vmware-tools-distrib# ./vmware-install.pl
```

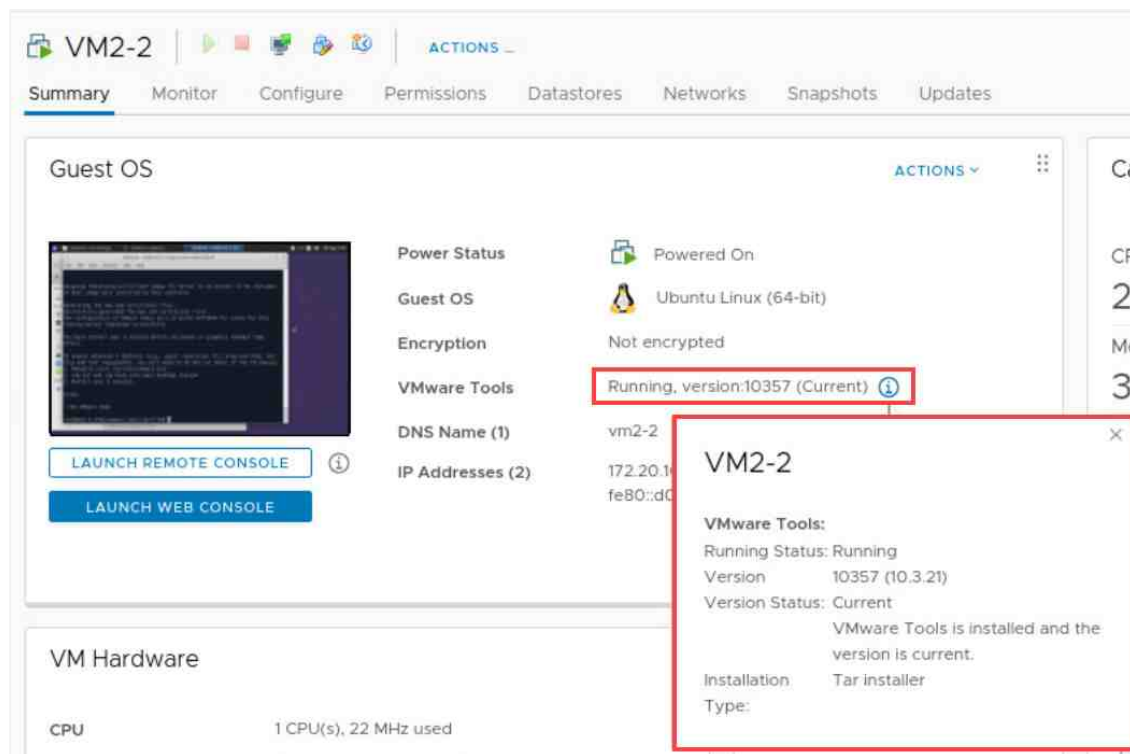
18. For this step, there will be a series of questions. Follow the instructions below to properly install the official supported version of VMware Tools.
- Do you still want to proceed with this installation? Enter `yes` and click Enter.
 - In which directory do you want to install the binary files? Click Enter.
 - What is the directory that contains the init directories (rc0.d to rc6.d)? Click Enter.
 - What is the directory that contains the init scripts? Click Enter.
 - In which directory do you want to install the daemon files? Click Enter.
 - In which directory do you want to install the library files? Click Enter.
 - The path `./usr/lib/vmware-tools` does not exist currently. This program is going to create it, including needed parent directories. Is this what you want? Enter `yes` and click Enter.
 - In which directory do you want to install the common agent library files? Click Enter.
 - In which directory do you want to install the common agent transient files? Click Enter.
 - In which directory do you want to install the documentation files? Click Enter.
 - The path `usr/share/doc/vmware-tools` does not exist currently. This program is going to create it, including needed parent directories. Is this what you want? Enter `yes` and click Enter.
 - Before running VMware Tools for the first time, you need to configure it by invoking the following command: `usr/share/vmware-config-tools.pl`. Do you want this program to invoke the command for you now? Enter `yes` and click Enter.
 - The VMware Host-guest Filesystem allows for shared folders between the host OS and the guest OS in a Fusion or Workstation virtual environment. Do you wish to enable this feature? Enter `No` and click Enter.

- n. The vmblock enables dragging or copying files between host and guest in a Fusion or Workstation virtual environment. Do you wish to enable this feature? Enter `no` and click Enter.
- o. Do you want to enable Guest Authentication (vgauth)? Enabling vgauth is needed if you want to enable Common Agent (caf). Enter `yes` and click Enter.
- p. Do you want to enable Common Agent (caf)? Enter `no` and click Enter.

19. Notice the message Enjoy, --the VMware team. This verifies the install of VMware Tools was successful.

```
Enjoy,
--the VMware team
root@vm2-2:/tmp/vmware-tools-distrib#
```

20. Change focus back to the vSphere Web Client. On the Summary pane of VM2-2, verify the officially supported version of VMware Tools is running. (10357)



Notice the version of VMware Tools is 10357 (10.3.21). This version is an officially supported version of VMware Tools. For lab purposes, if this pod had internet access, the version of open-vm-tools 11.1.0.19887 and build-16036546 would have been downloaded as an alternative to the officially supported VMware Tools. Some operating systems require the alternative if they are not officially supported by VMware.

4 Uninstall VMware Tools

In this task, you will uninstall VMware tools via the command line.

1. Change focus back to the VM2-2 console. In the terminal, type `exit`.

```
root@vm2-2:/tmp/vmware-tools-distrib# exit
```

```
root@vm2-2:/tmp/vmware-tools-distrib# exit
```

2. In the terminal, you will identify the VMware Tool package that is currently installed. Enter the command below, followed by pressing the Enter key.

```
sysadmin@vm2-2:~$ vmware-toolbox-cmd -v
```

```
sysadmin@vm2-2:~$ vmware-toolbox-cmd -v
10.3.21.249 (build-14772444)
sysadmin@vm2-2:~$
```



10.3.21.249 is the official supported version of VMware Tools installed in Task 3.

3. Elevate your privileges by becoming the superuser of the system. When prompted for the root password, enter `vmware123` and press Enter.

```
sysadmin@vm2-2:~$ sudo su
```

```
sysadmin@vm2-2:~$ sudo su
[sudo] password for sysadmin:
root@vm2-2:/home/sysadmin#
```

4. Change your current working directory to `/usr/bin`.

```
root@vm2-2:/home/sysadmin# cd /usr/bin/
```

```
root@vm2-2:/home/sysadmin# cd /usr/bin/
```



The `/usr/bin` directory is the location of the VMware Tools uninstaller script.

5. View the contents of the bin directory by typing the command below and press Enter. You will need to scroll the /usr/bin directory to find the vmware-uninstall-tools.pl script.

```
root@vm2-2:/usr/bin# ls
```

```
root@vm2-2:/usr/bin# ls
l                                nisdomainname                  symcryptrun
7z                               gettext                        sync
7za                             gettext.sh                     synclient
7zr                             ghostsript                     syndaemon
aa-enabled                      gigolo                         nm-connection-editor          system-config-printer
aa-exec                         gimp                          nm-online                    system-config-printer-applet
aconnect                       gimp-2.10                    nmtui                        systemctl
acpi_listen                    gimp-console                 nmtui-connect               systemd
add-apt-repository             gimp-console-2.10            nmtui-edit                  systemd-analyze
addpart                       gimp-test-clipboard-2.0      nmtui-hostname              systemd-ask-password
alsabat                       ginstall-info                nohup                       systemd-cat
alsaloop                      gio                           notify-send                  systemd-cgls
alsamixer                     gio-querymodules             nproc                       systemd-cgtop
alsatplg                      gipdddecode                  nroff                       systemd-delta
alsaucm                       glib-compile-schemas        nsenter                     systemd-detect-virt
amidi                         glxdemo                      nslookup                    systemd-escape
amixer                        glxgears                     nstat                       systemd-hwdb
aplay                         glxheads                     nsupdate                    systemd-idl28
aplaymidi                     glxinfo                      ntfs-3g                     systemd-inhibit
appport-bug                   gnome-font-viewer            ntfs-3g.probe               systemd-machine-id-setup
appport-cli                   gnome-help                   ntfscluster                  systemd-mount
appport-collect               gnome-keyring                 ntfscluster                  systemd-notify
```

```
File Edit View Terminal Tabs Help
chardet3                      i386                          plymouth                     uidparse
chardetect3                  i686-linux-gnu-pkg-config     pmap                         uxterm
chatrr                       iceauth                       pnm2ppa                      vdir
chcon                        ico                           pod2html                     vi
check-language-support       iconv                         pod2man                      view
chfn                         id                            pod2text                    views
chgrp                       iecset                       pod2usage                   vim.tiny
chmod                       ijs_pxljr                    podchecker                   vmhgfs-fuse
choom                       im-config                    podselect                    vmstat
chown                       im-launch                     poff                         vm-support
chrt                        info                          vmware-config-tools.pl      vmware-ctrl
chsh                        infobrowser                  POST                         vmware-guestproxycerttool
chvt                        infocmp                      ppdc                         vmware-hgfsclient
ciptool                     infotocap                    ppdhtml                     vmware-toolbox-cmd
ckbcomp                     inputattach                  ppdi                         vmware-uninstall-tools.pl
cksum                       install                       ppdmerge                    vmware-user
clear                       install-info                  pps                          vmware-xferlogs
cmp                         install-printrdriver          pr                           volname
codepage                    intel-virtual-output         precat                       vstp
col                          inxi                          preconv                      w
colcrt                      ionice                        preunzip                    wall
colormgr                    ip                             prezip                      watch
colrm                       ipcmk                        prezip-bin                  watchgnupg
column                      ipcrm                       print                        wc
comm                       ipcs                        printafm                    wdctl
compose                     IPPfind                      printenv                    wget
corelist                   iptool                      printerbanner               whatis
```

6. Run the VMware Tools uninstaller script by entering the command below and press the Enter Key.

```
root@vm2-2:/usr/bin# ./vmware-uninstall-tools.pl
```

```
root@vm2-2:/usr/bin# ./vmware-uninstall-tools.pl
```

7. Verify the removal of VMware Tools 10.3.21 was successful.

```
root@vm2-2:/usr/bin# ./vmware-uninstall-tools.pl
Uninstalling the tar installation of VMware Tools.

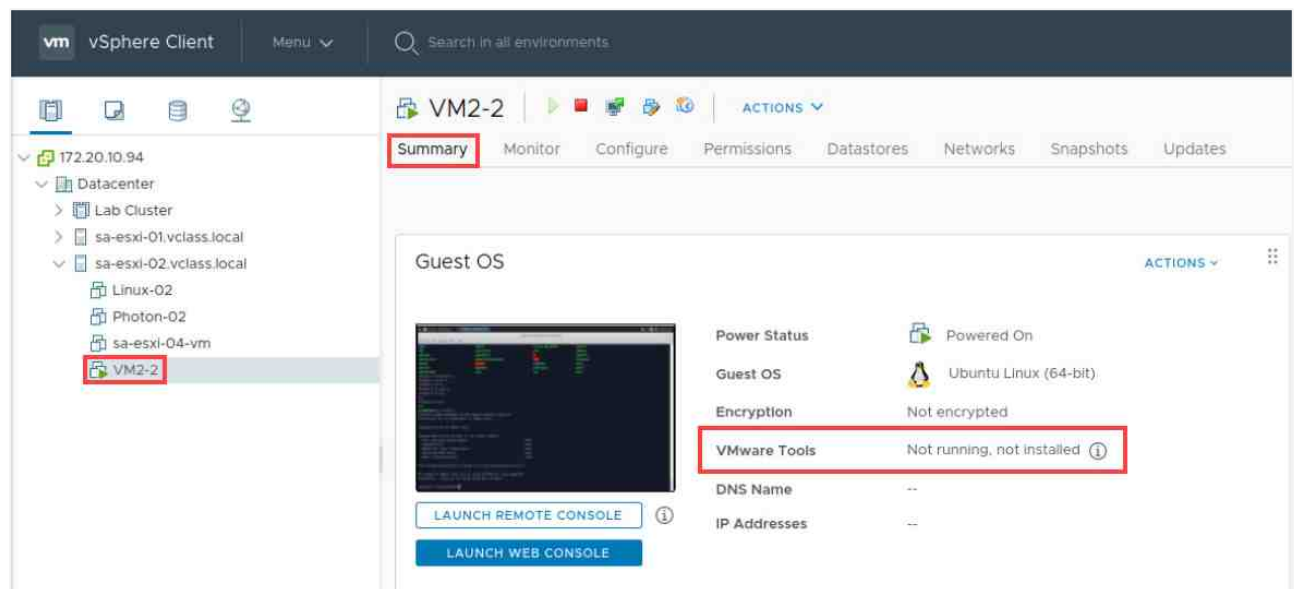
Stopping services for VMware Tools

Stopping VMware Tools services in the virtual machine:
  Guest operating system daemon: done
  VGAuthService: done
  VMware User Agent (vmware-user): done
  Unmounting HGFS shares: done
  Guest filesystem driver: done

File /etc/pulse/default.pa is backed up to /etc/pulse/default.pa.old.1.

The removal of VMware Tools 10.3.21 build-14772444 for Linux completed
successfully. Thank you for having tried this software.
```

8. Change focus to the vSphere Web Client. Click on VM2-2 from the navigator pane and click on the Summary tab in the main workspace. Notice that VMware Tools is not running and not installed.



9. The lab is now complete; you may end the reservation.